

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** CIVIL ENGINEERING

Commented [U1]: Please enter the name in Arial 12 font

2. **Subject Name:** Water and wastewater engineering **L-T-P:** 4-0-0
Credits: 4

Commented [U2]: Please enter the details (use Arial 12 font)

3. **Pre-requisites:** None

Commented [U3]: Please enter the pre-requisites (subject name and subject no. in Arial 12 font)

4. **Syllabus and reference books:**

Commented [U4]: 1. Please type the syllabus and reference books in the box provided below.

Syllabus:

Requirements for water supply systems; physical, chemical and biological characteristics of water; design of drinking water treatment plants and distribution systems; Generation and collection of wastewater; design of sewerage systems; primary, secondary and tertiary wastewater treatment systems; biological wastewater treatment systems; sludge disposal and handling; effluent disposal and wastewater reuse.

Reference Books:

- 1) Qasim, S., Motley, E., & Zhu, G. (2000). Water works engineering: planning, design and operation. Prentice Hall;
- 2) Metcalf & Eddy (2003). Wastewater Engineering: Treatment, disposal and reuse. New Delhi: Tata-McGraw Hill;
- 3) CPHEEO. (1999). Manual on water supply and treatment. Ministry of Urban Development, New Delhi;
- 4) Peavy, H., Rowe, D., & Tchobanoglous, G. (1985). Environmental Engineering. NY, US: McGraw Hill International Ed.

Please use "Arial" font with a font size of "12" and a line spacing of 1.0 for syllabus and reference books. Please use Justified Text (Ctrl+J) for the content entered in the boxes
Reference books should be numbered as 1), 2), 3)
Please provide all bibliographic details for the reference books

5. **Lecture-wise break-up:**

Commented [U5]: Please provide lecture-wise break-up in the table given below:

Sl. No.	Topic	No. of lectures
1.	General requirement for water supply, sources, quality and quantity of water, intake, pumping and transportation of water	4
2.	Physical, chemical and biological characteristics of water and their significance	2
3.	Engineered systems for water treatment: aeration, sedimentation, softening, coagulation, filtration, adsorption, ion exchange, and disinfection.	12
4.	Water distribution system	2
5	Generation and collection of wastewater, sanitary, storm and combined sewerage systems, Quantities of sanitary wastes and storm water.	4

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Commented [U6]:

	Design of sewerage system.	
6	Primary, secondary and tertiary treatment of wastewater; Wastewater disposal standards. Basics of microbiology.	4
7	Biological wastewater treatment systems: Aerobic processes - activated sludge process and its modifications, trickling filter, RBC, Anaerobic Processes- conventional anaerobic digester, High rate and hybrid anaerobic reactors.	12
8	Sludge digestion and handling. Disposal of effluent and sludge, onsite sewage treatment system and decentralized wastewater treatment system, Treated wastewater reuse.	8
Total number of hours		48

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